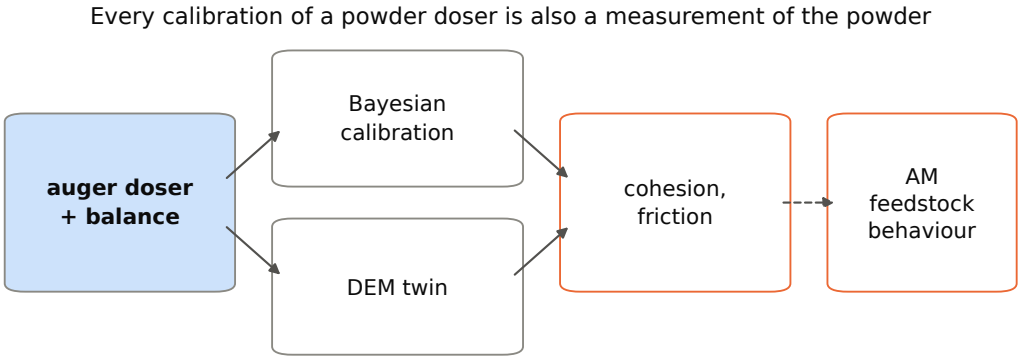
Graphical Abstract

Auger-based powder dosing as a probe of powder flow: multi-objective, multi-task Bayesian calibration and physics-based property inference

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Highlights

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- Calibration of an open-source auger doser is cast as Bayesian optimization
- Optimized settings dosed salt three times faster than hand-tuned settings
- Delivery per auger turn spans 24-fold across 13 powders at fixed settings
- Delivery ranks 12 powders like their literature angle of repose
- A DEM twin, not fitted to dosing data, matches salt delivery at one setting

Auger-based powder dosing as a probe of powder flow: multi-objective, multi-task Bayesian calibration and physics-based property inference

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Abstract

The mass an auger delivers depends on how it is driven and on the powder, so every calibration of an auger doser is also a small measurement of how that powder flows. We treat the calibration of an open-source, 3D-printed auger doser as multi-objective Bayesian optimization and ask what the calibration data reveal about the powder. At fixed settings, 13 powders, from food-safe stand-ins to AlSi10Mg and silicon, delivered between 10 and 231 mg per auger revolution, and three delivered none; across 12 of them, delivery ranked like the literature angle of repose of each material class (Spearman $\rho = -0.84$). On sodium chloride, an optimizer tuned eight controller settings in an unattended campaign of 42 doses, 14 of them chosen by the optimizer. Its recommended settings dosed 0.5 g in a median 78 s with a median error of 1.8 mg over eight repeats, three times faster than hand-tuned settings, and, in a single transfer check, dosed 8 g of AlSi10Mg to within 0.1% without change. We extend the formulation to many powders by treating each powder as a related task of a multi-task model, with added objectives for repeatability and accessible dose range. A discrete element model of the auger, with no parameters fitted to dosing data, reproduces the measured delivery of salt at the campaign's centre point (105 versus 105 ± 11 mg per revolution); with measured particle-size distributions, it is used to infer an effective cohesion

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and friction for each powder from its dosing response. [Multi-task calibration cut the doses needed for a new powder by X%, and the inferred parameters tracked shear-cell and Hall-flow measurements.] Because calibration is a step every dosing workflow performs anyway, its records can double, at little extra cost, as a characterization of each feedstock.

Keywords: powder dosing, powder flowability, Bayesian optimization, multi-task learning, discrete element method, self-driving laboratory

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1. Introduction

Alloy discovery by additive manufacturing (AM) starts with weighing powders. Each candidate composition is made by blending elemental or pre-alloyed powders, either in a bed for laser powder bed fusion (L-PBF) or in the feeders of a directed energy deposition (DED) system [1, 2, 3, 4, 5, 6]. In a self-driving laboratory (SDL), where a planning algorithm chooses each next composition and robots make and test it [7, 8, 9], that weighing step must be automatic, accurate, and fast for every powder in the palette [10, 11, 12, 13]. The same powders must later spread into thin, uniform layers, and how well they do so depends on the same properties that make them easy or hard to dose: particle size and shape, cohesion, and friction [14, 15, 16, 17].

Measuring those properties is slow and often uninformative. Hall and Carney funnels report a single flow time and cannot test strongly cohesive powders, which do not flow through them at all [18, 19]. Shear cells and powder rheometers give richer data, but each test needs tens of millilitres of powder, a dedicated instrument, and a trained operator [20, 21]. Rotating drums and spreadability tests come closer to the conditions in an AM machine [22, 23, 24, 25], yet different flow tests often rank the same powders differently, and none of them predicts spreading reliably on its own [18, 19, 24]. For an SDL that cycles through dozens of feedstocks, including small batches of reactive or expensive powders, a separate characterization campaign for each powder is rarely affordable.

Powder feeders, by contrast, are characterized all the time, because each new powder must be calibrated before it can be dosed. In pharmaceutical manufacturing, the feed factor of a screw feeder, the mass it delivers per screw revolution, has been predicted from bulk density, compressibility, and shear-cell data [26, 27, 28], and loss-in-weight feeders have standard methods for measuring how their output varies [29, 30, 31]. If material properties predict

feeder behaviour, the inverse should also hold: a feeder that is driven through a designed set of conditions, and weighed, measures the powder. Small-scale dosers make this attractive because the measurement uses milligrams to grams of powder and happens anyway, every time a new powder is loaded [32, 33, 34, 35, 36].

Calibration of this kind is an optimization problem with several competing goals. A dose should be accurate and repeatable, it should be fast, and the settings should work across the range of doses the workflow needs. Overshoot cannot be undone, because powder cannot be taken back out of the cup, so error in one direction costs more than in the other. Bayesian optimization (BO) suits such problems: it fits a probabilistic model to the doses made so far and chooses the next settings where the expected gain is largest [37, 38, 39], and it has already tuned additive manufacturing processes autonomously [40]. Multi-objective BO finds the trade-off, or Pareto front, between goals such as time and error [41], and sparse priors make it practical with many settings and few experiments [42]. Each new powder, however, still starts from scratch. Multi-task BO treats related problems, here different powders on the same doser, as tasks of one model, so that data from earlier powders inform the first doses of a new one [43, 44, 45]. How strongly two powders are related is then a learned quantity, and it is natural to ask whether it reflects their physical properties.

Physics-based models can answer that question directly. The discrete element method (DEM) simulates every grain and its contacts [46, 47], has been used to predict the output of screw conveyors [48], and represents cohesion and rolling friction with contact laws whose parameters are hard to measure [49, 50, 51]. These parameters are usually calibrated by matching simulations to bulk tests, which is itself an inverse problem [52, 53, 54, 55], best treated with the tools of Bayesian calibration of computer models [56, 57]. The closest precedent to the present work calibrated DEM parameters of pharmaceutical powders against several bulk responses extracted from auger dosing experiments [58, 59]; here the same idea is applied to an instrument that also optimizes its own settings, and to AM feedstocks. DEM is also the standard tool for studying how powders spread into AM layers [60, 61, 15, 62]. An auger that is turned at controlled speeds and tilts is a well-defined shear and discharge test, so a DEM model of the auger could convert dosing data into contact parameters that a spreading model can then use.

In this work we build on an open-source, 3D-printed auger doser [63] and make three contributions. First, we formulate per-powder calibration

as multi-objective BO over the doser's controller settings and demonstrate it on sodium chloride, where an eight-dimensional search ran unattended (Section 3.2). Second, we show that the doser's response at fixed settings separates 13 powders over a 24-fold range, and we extend calibration to many powders as a multi-task problem whose objectives also include repeatability and the range of doses a powder can reach (Sections 3.1 and 3.3). Third, we build a DEM model of the auger, validate it against measured delivery without calibration, and use it to infer effective cohesion and friction from dosing data, which we compare with Hall-flow and shear-cell measurements (Sections 3.4 and 3.5). All hardware, firmware, optimization code, simulation inputs, and data are open [64].

2. Materials and methods

2.1. Doser and dose procedure

The doser is a single channel of an open-source design described in detail elsewhere [63, 64]; the parts that matter here are summarized below. A one-piece auger, printed in PLA on a Bambu Lab H2D printer (0.2 mm layers), serves as both reservoir and screw: a tube of 21 mm bore and 250 mm length holds up to 58 mL of powder, and a single-start flight (0.5 mm thick, 10.4 mm pitch, 8 turns) runs over the last 84 mm before a 12 mm conical outlet with a 3 mm exit hole. A NEMA-11 stepper motor (11HS18-0674S, Pololu Tic T500 driver) turns the auger through a 20:44 gear pair, up to about 109 rpm. Two servos tilt the auger about a hinge near the outlet through a 2:1 gear, and a push–pull solenoid (JF-0530B, 60 ms pulses) taps the tube to free powder held at the lip. Powder falls into a cup on an analytical balance (A&D HR-100A; 0.1 mg readability, 102 g capacity) that reports over RS-232. A Raspberry Pi Pico W runs the dosing firmware (MicroPython), and a Raspberry Pi Zero 2 W next to the rig runs each dose, records it, and uploads it to a MongoDB database. All tilts below are angles of the auger axis above horizontal.

A dose has three stages (Fig. 1a). In the *bulk* stage the auger turns continuously at a set tilt and speed, optionally while the solenoid taps at 2 Hz, until the mass still to go falls to a switch threshold plus a fixed 50 mg allowance for powder still in flight. In the *trickle* stage the auger runs slowly (typically 2–15 rpm) under a proportional–integral controller on the flow rate, which a Kalman filter estimates from the balance readings. The trickle stops when the predicted final mass, the current estimate plus the expected

afterflow (the flow rate times an afterflow time constant τ_{af}) plus one standard deviation of the estimate, comes within a fixed 35 mg margin of the target. In the *tap* stage the auger stops, the tube moves to a third tilt, and single taps free a few milligrams at a time until the reading lies within a stopping band of $\pm\delta$ around the target. The dose is scored on a balance reading taken 2 s after the last action. Doses that exceed the target by more than 100 mg are aborted, and doses that use up their tap budget end as jams. The Kalman filter models the balance as a first-order lag; the controller used the datasheet value of 0.7 s, whereas drop tests on the rig gave 0.16 s, a mismatch that the optimized settings absorb but that should be removed before settings are compared across powders. Every dose writes one machine-readable record with the settings as executed, the time spent in each stage, every stop event, and the balance telemetry at 4 Hz.

2.2. Powders

We report dosing data for 13 powders (Table 1). Seven are food-safe stand-ins chosen to span flow behaviour (table salt, calcium lactate, xanthan gum, carboxymethyl cellulose, white and brown rice flour, sodium alginate), and six are relevant to alloy and materials work (gas-atomized AlSi10Mg, silicon sieved to 75–150 µm and to below 44 µm, sodium sulfate, barium chloride dihydrate, and fumed silica). An Al 4047 powder made in our laboratory by ultrasonic atomization was also dosed. Particle-size distributions, densities and flow properties have not yet been measured on these lots. For the correlation analysis in Section 3.5 we therefore use literature and supplier values for the material class, with each value traced to its source in the repository [64]; seven of the 12 angles of repose are engineering estimates. [Measured properties of each lot (Section 2.7) will replace these values.]

2.3. Gravimetric response at fixed settings

Each powder was first run through a fixed battery of test protocols, described fully with the hardware [63]. The ones used here are protocol C, six single auger revolutions at 30 rpm at each of three tilts (0, 22.5 and 45°), each followed by a settled balance reading; protocol D, three revolutions at 15, 45 and 90 rpm at 22.5°; protocol E, single taps after a re-feed revolution; and protocol G, three closed-loop doses of 1 g with an earlier, simpler three-stage controller whose settings were tuned by hand on salt and frozen for every powder. From these we extract a short *dosing fingerprint* for each powder (Table 2): the delivery per revolution at the three tilts, its scatter from

Table 1: Powders and their response to the auger at fixed settings (protocol C: 30 rpm, six single revolutions per tilt). Delivery is the mean mass per auger revolution ± standard error at 22.5° tilt. Closed-loop doses used an earlier three-stage controller with one set of settings tuned by hand on salt. "–" marks powders that delivered no measurable amount.

Powder	Group	mg/rev at 22.5°	1 g doses within ±5%
AlSi10Mg (gas-atomized)	research	231.0 ± 12.7	3/3
Silicon, 75–150 µm	research	210.7 ± 6.0	3/3
Sodium sulfate	research	208.4 ± 13.1	3/3
Calcium lactate	stand-in	198.3 ± 1.8	3/3
Barium chloride dihydrate	research	184.7 ± 12.9	not dosed (caked)
Xanthan gum	stand-in	161.2 ± 7.9	3/3
Sodium chloride (table salt)	stand-in	146.5 ± 11.0	6/6
Carboxymethyl cellulose	stand-in	26.4 ± 1.0	3/3
White rice flour	stand-in	12.8 ± 2.0	0/3 (short)
Sodium alginate	stand-in	9.6 ± 0.3	0/3 (short)
Silicon, <44 µm	research	–	0/6
Brown rice flour	stand-in	–	0/3
Fumed silica	research	–	not dosed

revolution to revolution, the exponent α in $\dot{m} \propto N^{\alpha}$ for auger speed N, the mass freed by a single tap, and the afterflow time constant τ_{af} fitted from the mass that arrives after each stop. A full battery uses 4–10 g of powder and takes 40–50 min. Two further descriptors were proposed for screening atomized batches [64]: a gravity-assisted volumetric flow, the delivery per revolution at 45° divided by bulk density and normalized to a reference AlSi10Mg powder, and an interruption fraction, the share of balance increments at 15 rpm that show no flow, which is a doser analogue of the cohesive index of a rotating drum [65, 22].

2.4. Calibration as multi-objective Bayesian optimization

For one powder and one target mass, calibration searches eight controller settings (Table 3) to minimize two objectives: the total time of a dose, t, and its absolute error, $|e|$, both measured on the scored balance reading. The campaign fixed no tolerance in advance: the stopping band δ is itself one of the searched settings, and the error objective measures where the dose actually ended. Doses slower than 180 s or more than 20 mg off count as failures; these limits form the reference point for the hypervolume, the area

of objective space that the best doses so far dominate. Each dose ends in one of a fixed set of outcomes: *ok* (within the band), *overshoot* (above the band), *jam* (tap-cycle or nudge budget used up, or no progress for 8 s), *timeout* (600 s), *abort* (100 mg above the target), or an infrastructure error (balance or communication), which is flagged separately; spills are flagged by an operator and were therefore not recorded in unattended runs. All other controller parameters, including the controller gains and the overshoot guard, were frozen at the values used in the hand-tuned controller.

The campaign ran in three phases, with four doses at the hand-tuned settings spread through it as a baseline and a check on drift. First, a two-level fractional factorial screening design of resolution IV (2^{8-4}, 16 corners) estimated main effects without confounding them with two-factor interactions, and four repeated centre points measured the dose-to-dose scatter. Second, the afterflow constant τ_{af} was refitted from the 26 stop events recorded so far (a robust median-ratio estimate; $\tau_{\mathrm{af}} = 0.834$ s), and the centre point and hand-tuned settings were repeated with the new value. Third, 14 doses were chosen sequentially in Ax 0.4.3 [66] on BoTorch [67]: two quasi-random (Sobol) doses, then 12 chosen by a fully Bayesian Gaussian-process model with a sparse axis-aligned subspace prior (SAASBO) [42], suited to many settings and few observations, and the logarithmic form of the noisy expected hypervolume improvement (qLogNEHVI) [41, 68], which accounts for the noise in each observed dose. Screening and centre-point doses were attached to the experiment as existing data so that the model used all 28 earlier doses. Doses that jammed were reported to the model at the failure limits (180 s, 20 mg), and overshoots at their measured values. Each suggestion took 1–1.5 min to compute. The optimization phase was stopped by a preset deadline after 14 doses, short of the 30–40 planned. Each dose was run unattended over a secure shell connection to the rig, with the cup emptied at fixed intervals; a restart never repeated a dose, because each dose carries a unique identifier and its stored result is returned instead. The campaign and the validation blocks were launched and monitored by an AI coding agent (Claude Code, Anthropic), which also wrote much of the firmware, campaign, analysis and simulation code under review by the authors. The agent started the campaign script, watched it and restarted it after interruptions; the settings of every dose came from the screening design or the optimizer, not from the agent. Every prompt and response is public in the repository [64].

Because the observed front is built from single doses, a point can lie on it by chance. We therefore recommend settings from the model's Pareto set

(posterior means) rather than from the observed front, and validate them with a block of eight repeated doses on a later day. The validation block also tested the screening corner that lay on the observed front (`corner-09`).

2.5. Multi-task calibration across powders

Each powder is a task, and all tasks share the eight controller settings and the dose record. A multi-task Gaussian process [43, 44] models the objectives of task k at settings $\mathbf{x}$ with a product kernel $K\Big((\mathbf{x}, k), (\mathbf{x}', k')\Big) = B_{kk'}\, k_x(\mathbf{x}, \mathbf{x}')$, where the positive semi-definite matrix B is learned from the data and its normalized entries are the correlations between powders. Two variants, both available in the pinned version of Ax, will be compared: an index kernel, which learns B freely, and a descriptor kernel, in which B is computed from each powder's dosing fingerprint (Section 2.3) and, where available, its measured properties, so that a new powder can borrow from similar ones before its first optimized dose [45]. The objectives are extended from two to four, following the formulation developed for this controller [64]: (i) accuracy, the median absolute error; (ii) repeatability, the probability that a dose lands within tolerance, $P(|e| \leq \delta)$, estimated from replicate blocks, together with a constraint on the probability of overshoot, which cannot be undone; (iii) time per dose; and (iv) the accessible dose range, the smallest target that the settings deliver within tolerance with probability at least 0.95. [Final formulation (constraint thresholds, replicate count per evaluation, and dose targets) to be fixed before the multi-task campaigns.]

The benchmark has two parts. Retrospectively, each completed single-task campaign is held out in turn, a surrogate fitted to its data stands in for the rig, and multi-task optimization on the held-out powder, informed by the other powders, is compared with single-task SAASBO, quasi-random (Sobol) search, and simply carrying over the best settings of the most similar powder. Prospectively, one new powder, a freshly atomized batch, is calibrated with the multi-task model. The figure of merit is the number of doses until the recommended settings reach 90% of the best hypervolume known for that powder. [Planned campaigns: AlSi10Mg, 316L stainless steel, sieved Al 4047 and at least one elemental alloy precursor, each with about 42 doses and an 8-dose validation block.] Reactive elemental precursors will be dosed in an argon glove box; where the elemental form is too hazardous to handle as a fine powder, a master alloy will be used instead.

2.6. Discrete element model of the auger

A discrete element model of the auger outlet was built in LIGGGHTS-PUBLIC 3.8 [69, 46]. Grains interact through a Hertz normal contact with history-dependent tangential friction and an elastic–plastic spring–dashpot rolling resistance model [51], and cohesion is added as a simplified Johnson–Kendall–Roberts (SJKR) term, a pull-off force proportional to the cohesion energy density and the contact area [49]. For salt, the grains follow a three-size distribution around a median diameter of 0.425 mm (number fractions 0.3/0.4/0.3 at 0.85, 1 and 1.15 times the median) with density 2165 $\mathrm{kg\,m^{-3}}$, a literature value for the material class. Young's modulus is reduced to 0.2 MPa to allow a time step of 2.1×10^{-5} s, which keeps mean overlaps below 0.4% of a diameter; Poisson's ratio is 0.3, the coefficient of restitution 0.5, sliding friction 0.5 between grains and 0.4 against the wall, and rolling friction 0.3. None of these parameters was fitted to the dosing data. The geometry is meshed from the CAD model of the printed auger: the conical outlet with its 3 mm exit, the solid core whose tip leaves a 1.07 mm annular gap at the exit, and the lowest 1.5 turns of the flight, filled with about 29,000 grains and topped up from a feed zone. Tilt is applied by rotating gravity, rotation by turning the auger mesh, and the delivered mass is counted as grains leave the exit. Each case was run for 0.3–3 revolutions, mostly from a single random packing. Because the SJKR pull-off force scales with the contact area, which depends on the softened stiffness, any cohesion energy density inferred with this model is a simulator-effective parameter, comparable across powders only at fixed numerical stiffness; particle–wall adhesion to the printed PLA is a separate parameter that dosing data alone may not resolve.

To infer effective contact parameters for a powder, the model will be run over a designed set of cohesion energy densities, sliding and rolling friction coefficients, with the particle-size distribution fixed at the measured one, at the tilts and speeds of the dosing fingerprint. A Gaussian-process emulator of these runs then replaces the simulator in a Bayesian calibration [56, 57], which returns a posterior distribution over the contact parameters given the measured fingerprint and its scatter, with a discrepancy term for systematic model error. Because dosing data alone may not separate cohesion from friction [52, 53, 59], the identifiability of each parameter will be checked on synthetic data before the inference is run on measured data, and grain counts for the finer metal powders will be reduced by coarse-graining that preserves the ratio of cohesive to gravitational forces [70]. [Emulator design (number of runs, parameter ranges) to be fixed after the friction and cohesion sweeps in the

rig geometry.]

2.7. Independent powder characterization

To test whether the dosing-derived descriptors and inferred parameters carry physical meaning, the same lots will be characterized by standard methods [71]: particle-size distribution by laser diffraction [72] or sieving, bulk and tapped density and the Hausner ratio [73], Hall-flow time [74] or, for powders that do not pass the Hall funnel, Carney-flow time [75], angle of repose, and the flow function coefficient ff_c from a shear cell [76, 20]. Spreadability will be assessed by blade-spreading single layers of 50–100 µm and measuring areal mass and coverage [24, 25]. [Shear-cell measurements are expected to be outsourced; sieves have been ordered.]

2.8. Statistical analysis

Delivery per revolution is reported as the mean ± standard error of single revolutions, and a value counts as a measurement only when the mean exceeds twice its standard error. Associations between dosing descriptors and powder properties are reported as Spearman rank correlations, because $n \leq 12$ and many property values are class estimates; p-values are not corrected for the 42 predictor–response pairs examined and are treated as leads. The optimization models are specified completely by the pinned software versions and the campaign configuration stored with each campaign, and all trials, including aborted and jammed doses, are kept in the analysis. [For the multi-task and inference results, report baselines, held-out performance, residuals and hyperparameters as required by the journal's guidelines for machine-learning studies.]

3. Results

3.1. Dosing response separates powders

At fixed settings the auger delivered very different amounts of different powders (Fig. 2a, Table 1). Ten of the 13 powders delivered between 9.6 mg (sodium alginate) and 231 mg (AlSi10Mg) per revolution at 22.5°, a 24-fold range, and three delivered no measurable amount: fumed silica, brown rice flour, and silicon finer than 44 µm. The two silicon grades are the clearest case. They are the same material, yet the 75–150 µm grade delivered 211 mg per revolution and the finer grade delivered nothing in 17 of 18 revolutions, so particle size alone switched the flow on and off.

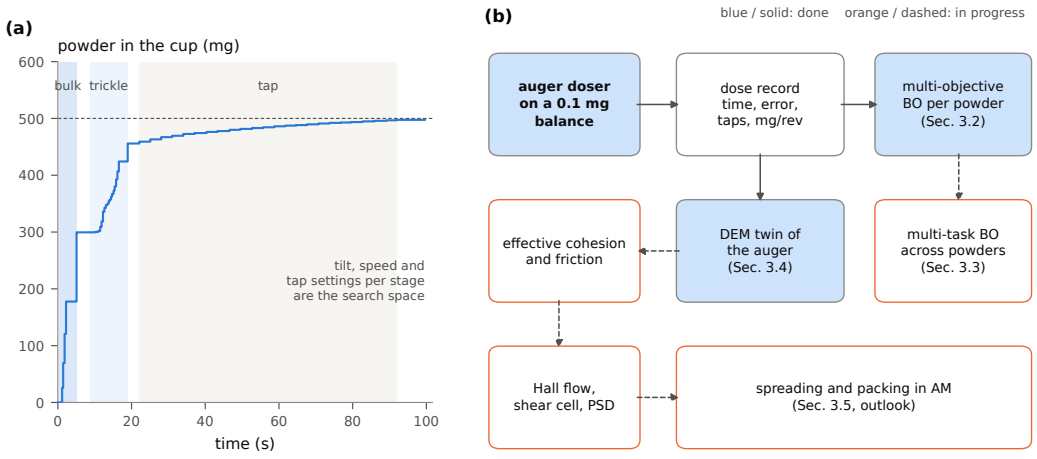


Figure 1: Calibration of an auger doser as a measurement of the powder. (a) One 0.5 g dose of salt at the recommended settings, as the balance recorded it: a fast bulk stage, a slow trickle stage and a tap stage, each with its own tilt, speed and tap settings. (b) The data flow of this work. Dose records feed a multi-objective optimizer for each powder and a multi-task model across powders; a discrete element model of the auger converts the same records into effective contact parameters, which are compared with standard flow tests and, in future work, with spreading in additive manufacturing. Solid boxes and arrows mark completed work; orange outlines and dashed arrows mark work in progress.

Across all powders and tilts, powder identity accounted for 70% of the variance of the logarithm of the mass per revolution, tilt for 20%, their interaction for 4%, and repeat-to-repeat scatter for 6%. Tilting the auger from 0 to 22.5° raised delivery by a median factor of 4.2, and tilting further to 45° by only 1.2 more; carboxymethyl cellulose was the only powder that delivered less at 45° than at 22.5°. Delivery rose more slowly than auger speed for most powders (median exponent $\alpha = 0.78$), but faster than speed for the three slowest (white rice flour 1.36, carboxymethyl cellulose 1.13, sodium alginate 1.08). Put another way, free-flowing powders delivered less per revolution as the auger sped up (salt by 33–47%, AlSi10Mg by 49% over a sixfold range of speed), whereas cohesive ones delivered more (white rice flour 94% more, carboxymethyl cellulose 23% more), so the sign of the speed response alone separates the two groups. A single tap freed between 0.24 mg (sodium alginate) and 20 mg (calcium lactate), and the afterflow time constant ranged from 0.78 to 1.17 s across six powders. Each of these numbers comes from data that any calibration of the doser collects anyway.

The same powders also differ in how well a fixed controller doses them

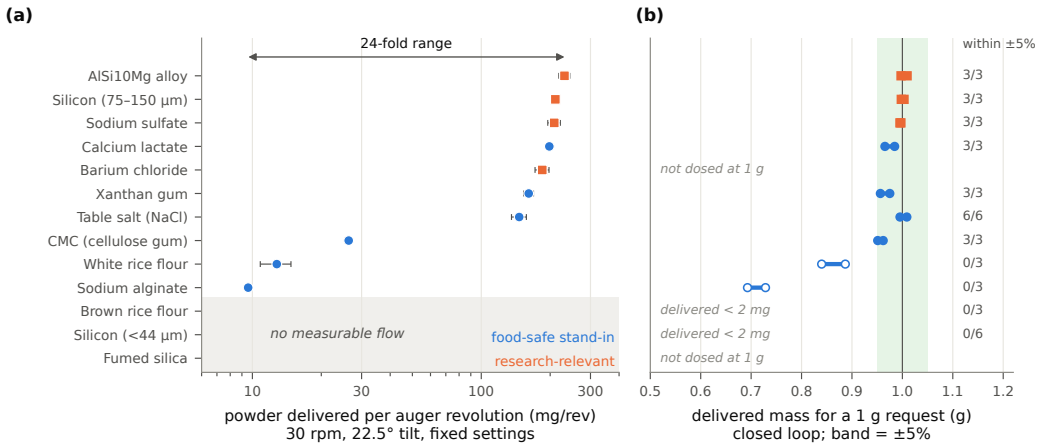


Figure 2: Thirteen powders on one auger. (a) Mass delivered per auger revolution at fixed settings (30 rpm, 22.5° tilt; mean ± standard error of six revolutions). Three powders delivered no measurable amount. (b) Delivered mass for 1 g requests under closed-loop control with one set of hand-tuned settings; bars span the replicate doses, open markers show powders that missed the ±5% band, and the numbers give doses within the band. Data from the companion hardware study [63].

(Fig. 2b). With an earlier three-stage controller and one set of settings tuned by hand on salt, seven powders landed within ±5% of a 1 g target on every dose (24 of 24), and the three alloy-relevant ones (AlSi10Mg, coarse silicon and sodium sulfate) within 1%. The two slowest-flowing powders ran out of time: white rice flour delivered 0.84–0.89 g and sodium alginate 0.69–0.73 g. Even among powders that met the limit, the median time for 1 g ranged from 65 s for AlSi10Mg to 11 min for carboxymethyl cellulose. A controller tuned for one powder is therefore accurate on similar powders but slow or short on different ones, which is the case for calibrating each powder.

3.2. Bayesian calibration on salt

The salt campaign ran 42 doses of 0.5 g unattended in less than 2.5 h (Fig. 3). The four doses at the hand-tuned settings took 180–343 s (median 245 s) and ended 3.6–4.8 mg short. The screening design showed which settings mattered and why. Tapping during the bulk stage cut the average time of successful doses from 186 to 79 s, but five of the eight corners with bulk taps on ended in an overshoot or a jam, against two of the eight with taps off, presumably because taps free extra powder after the auger stops. Raising the tap-stage tilt from 0 to 15° cut the average time from 223 to 92 s.

Switching to the trickle stage earlier, with 0.30 g rather than 0.05 g to go, cut the mean absolute error from 34 to 11 mg, and two corners at the larger threshold ended in an overshoot or a jam, against five at the smaller. In total, 8 of the 42 doses overshot, by 21–83 mg, six of them during screening and every one at the highest bulk speed, the smallest switch threshold, or both; three more ended in a jam when the tap budget ran out.

The optimizer then chose 14 doses before the campaign reached its deadline. None of them beat the best single doses of the screening design, so the observed hypervolume did not grow during this phase. The settings the model recommended (`bo-005`, Table 3) had been predicted to take 91 ± 66 s with a 3 ± 9 mg error, and took 100 s with a 2.4 mg error in the campaign. Repeated eight times a week later, they dosed 0.5 g in a median 78 s (range 60–114 s; bootstrap 95% interval of the median 69–110 s) with a median error of 1.8 mg (1.3–2.3 mg; all doses 0.8–2.6 mg under the target) and no overshoot (Fig. 3b); the delivered masses had a standard deviation of 0.56 mg. Eight successes in eight doses bound the probability of a dose inside the band only from below at 0.69, and no overshoot in eight bounds its probability from above at 0.31 (one-sided 95% Clopper–Pearson limits); demonstrating a success rate of 0.95 would take 59 doses without a failure. That is three times faster than the hand-tuned settings with less than half their error. The fastest single dose of the campaign, the screening corner `corner-09` at 52 s and 0.7 mg, did not repeat: eight more doses at its settings took 47–385 s (median 202 s), depending on how much powder happened to fall after the bulk stage stopped (22–69 mg). The observed Pareto front therefore overstated what that setting could do, and a recommendation based on the model's averaged prediction held up where the single best dose did not.

Most of the time saved came from the tap stage (Fig. 3c). A hand-tuned dose spent 152–318 s, 84–93% of its time, tapping out the last 54–66 mg at 0.6–1.1 mg per tap, whereas at the recommended settings each tap freed 1.5–3.6 mg, a difference consistent with the tube rising from a 10° trickle tilt to a 15° tap tilt instead of dropping to it. The optimizer changed seven of the eight settings, and all six numeric settings ended at an edge of their search range (Fig. 3d). Some of these edges are hardware limits, such as the auger speed near its maximum, and others are choices, such as the 3 mg stopping band; the rest suggest that the next campaign should search a wider range.

The salt settings also transferred to a different powder. Applied unchanged to AlSi10Mg, with the afterflow constant fitted on salt, they delivered 7.993 g for an 8 g request (−0.09%) in 315 s, in a single dose that has not yet been

repeated. An unsieved Al 4047 batch from our atomizer, in contrast, clogged the auger on coarse agglomerates. Free-flowing powders thus appear to share much of their calibration, while powders that flow differently need their own, which is what a multi-task model is designed to exploit.

3.3. Multi-task calibration across powders

[This subsection reports results that do not exist yet; the text below shows the intended form.] With data from [three] earlier powders, the multi-task model reached 90% of the best known hypervolume on a held-out powder in [N_{MT}] doses, against [N_{ST}] for single-task optimization and [N_{Sobol}] for quasi-random search (Fig. 4a). Carrying over the best settings of the most similar powder gave a good first dose for free-flowing powders but [failed for cohesive ones]. The learned correlations between powders (Fig. 4b) [grouped the free-flowing metal and salt powders together and separated the cohesive and hygroscopic ones], and the descriptor kernel built from dosing fingerprints [matched the index kernel after one dose on the new powder]. In the prospective test on a freshly atomized batch, the recommended settings [dosed 0.5 g in X s with Y mg median error after Z doses]. Across the accessible-dose-range objective, the smallest target dosed within tolerance with probability 0.95 was [50 mg for AlSi10Mg and 200 mg for salt].

3.4. A discrete element twin of the auger

Without any fitted parameter, the discrete element model reproduced the delivery of salt at the campaign's centre-point settings: 105 mg per revolution at 27.5° and 60 rpm averaged over the run (116 ± 20 mg after the first quarter revolution), against 105 ± 11 mg measured over eight doses (Fig. 5a). A second random packing at the same settings gave 102 mg. The model also reproduced the rise of delivery with tilt, but with larger residuals at the other settings: 33 against about 28 mg per revolution at 0° (+18%), 153 against about 195 mg at 45° (−22%), and 71 against 106–110 mg at 22.5° and 90 rpm (−33 to −35%). The measured values at 0° and 45° were taken at 30 rpm and scaled to 60 rpm with the measured speed exponent.

The model is most sensitive to the geometry of the exit and to the grains. The tip of the auger core sits inside the exit and leaves a 1.07 mm annular gap, about 2.5 grain diameters, which throttles the flow: moving the tip back by 0.5, 1 and 2 mm raised delivery from 105 to 112, 224 and 427 mg per revolution. Reducing the median grain size from 0.425 to 0.35 mm raised it by 29%. In a smaller 10 mm auger, adding cohesion (20 kJ m^{-3}) cut delivery

from 46.8 to 0.2 mg per revolution, the same switch from flow to no flow seen between the two silicon grades. The model's cost sets what can be simulated: the outlet section with 29,000 grains takes 26–30 min per revolution on one processor core, a full salt-filled auger with 1.1 million grains about 19 h, and a fine metal powder would need about 10^9 grains per auger, so those powders require coarse-graining [70].

[Inference results.] The emulator built from [N] simulations reproduced held-out simulations within [X%]. Delivery depended most strongly on cohesion at low tilt and on friction at high tilt (Fig. 5b), so the tilt sweep of the dosing fingerprint constrains both. For salt the posterior gave an effective cohesion energy density of [X kJ m^{-3}] and sliding friction of [Y] (Fig. 5c); for [the other powders] the values are listed in Table [S1].

3.5. Dosing response and powder properties

Even with literature values in place of measured properties, the dosing response ranked powders the way standard flow measures do (Fig. 6a). Across 12 powders, delivery at 45° fell as the angle of repose rose (Spearman $\rho = -0.84$, bootstrap 95% interval -0.98 to -0.50; $p = 0.0006$, or 0.023 after a Bonferroni correction for the 42 pairs examined), and the same held at 22.5° ($\rho = -0.77$, $n = 11$). The five powders with a measured or manufacturer angle of repose ranked correctly on their own: those at 30–32° delivered 244–339 mg per revolution and those at 41–45° delivered 9–37 mg. The equilibrium moisture content predicted the error of 1 g doses under the hand-tuned controller ($\rho = +0.87$, $n = 9$): every powder dosed within 5 mg held less than 0.1 wt% water, and every hygroscopic biopolymer missed by 33–292 mg. Literature median particle size did not predict delivery across materials ($\rho = +0.26$), although it decided flow within one material, as the silicon grades show. The shape of the tilt response also differed between powders (Fig. 6b), which is the information the discrete element model needs to separate cohesion from friction. The interruption fraction, computed from the same balance records, ranged from 0.13 for AlSi10Mg and 0.26 for salt to 0.81 for carboxymethyl cellulose and 0.96 for sodium alginate, and it also followed the angle of repose ($\rho = +0.76$, $n = 10$) while only partly overlapping the volumetric flow index ($\rho = -0.66$ between the two), so the two descriptors are not redundant.

[Comparison with measured properties.] On the same lots, the effective cohesion inferred from dosing correlated with the shear-cell flow function coefficient ($\rho =$ [X], Fig. 6c) and with the Hall-flow time of the powders that

passed the funnel ([Y]); [Z] powders did not pass the Hall funnel, but all of them produced a dosing fingerprint. Against blade-spread single layers, [the interruption fraction ranked the batches like layer coverage].

4. Discussion

Calibration data as a flow measurement. The doser measures a powder under conditions close to those of its use: a small sample, moving through a narrow gap under gravity, shear and vibration. Every quantity in the dosing fingerprint is collected during routine calibration, and the same measurement works inside a glove box, where Hall funnels and shear cells are awkward to use. A full battery takes 4–10 g and under an hour; when the calibration has to be done anyway, storing and analyzing its records adds little. The fingerprint has two further advantages over funnel tests. It degrades gracefully on cohesive powders: where nothing moves, a defined protocol returns a censored no-flow result, and wherever powder moves at all it returns tilt, tap and afterflow descriptors, whereas a funnel that does not flow returns only a failure [18, 77]. And it has several dimensions, so it can separate effects that a single flow time combines. The correlation with the angle of repose and the failure of particle size to predict delivery across materials (Section 3.5) agree with studies of pharmaceutical screw feeders, in which bulk density and cohesion-related properties, rather than size, predicted feed rate [26, 27, 28, 78]. The literature values used here are class estimates, however, not measurements of our lots, and a correlation of ranks across 12 very different materials shows only that the doser orders powders sensibly. Whether it can resolve the smaller differences between batches of one alloy, which matter most for feedstock quality control in AM, remains to be shown [(Section 3.5)].

What Bayesian optimization contributes. On salt, the optimizer found settings three times faster than hand tuning with less error, in one unattended session of under 2.5 h. Three lessons from the campaign shape the multi-powder design. First, repeatability has to be an objective, not an afterthought. The fastest dose on the observed front came from settings whose time varied eightfold between repeats, because the powder that falls after the auger stops varies from dose to dose. A noise-aware acquisition function [41], a recommendation based on the model's averaged prediction, and a validation block together guarded against this, but an explicit repeatability objective, estimated from replicate doses, would let the optimizer trade it against speed

directly. Second, a short budget finds a good setting but does not prove it best: the 14 optimizer doses never beat the luckiest screening doses, and only the validation block showed which setting was both fast and repeatable. Time per dose was heavy-tailed (the eight doses at the centre point took 24–391 s), so a noise model that allows for occasional very slow doses, and a longer budget, would sharpen the comparison. Third, the recommended settings sat at the edges of the search space, so the ranges were set too conservatively, which is easy to fix but should be expected for every new powder. The screening design was worth its 20 doses: it explained, in physical terms, why taps during the bulk stage speed doses up but cause overshoots, and it gave the model a balanced starting set. Because overshoot is irreversible, the current formulation's symmetric absolute error should become a one-sided penalty with a constraint on the probability of overshoot, as the formulation for the four-objective campaigns does (Section 2.5).

Sharing calibration across powders. That the salt settings dosed AlSi10Mg to within 0.1%, in a single transfer check, suggests that powders which flow freely share much of their calibration, and that a multi-task model can start a new free-flowing powder close to its optimum. Cohesive and hygroscopic powders, which deliver an order of magnitude less per revolution and respond differently to tilt and speed, are less likely to benefit from the salt data, and a model that learns task correlations decides this from data rather than by assumption. Using the dosing fingerprint to compute the task correlations would also let the model place a new powder before its first optimized dose [45]. [Results of the multi-task benchmark (Section 3.3) to be discussed here.]

From dosing data to contact parameters. The discrete element model reproduced salt delivery at one setting without calibration and followed its trend with tilt, which suggests that its geometry and contact laws capture the main mechanism. Its errors at other settings, and its strong sensitivity to the exit gap and the grain size, mean that inference must start from a measured exit geometry and a measured particle-size distribution, or those uncertainties will be absorbed into the inferred cohesion and friction [52, 59]. The inferred parameters will also be effective rather than intrinsic: with a softened Young's modulus, the cohesive force in the SJKR model depends on the contact area and hence on the chosen stiffness, so inferred cohesion energies can be compared across powders only at a fixed numerical stiffness [50]. Triboelectric charging, suspected in the noisy readings of carboxymethyl cellulose and seen

as salt sticking to the auger walls in a glove box, is not represented in the model [79]. If these caveats are respected, the same contact parameters can drive DEM models of powder spreading, which are the standard tool for predicting layer quality in powder bed fusion [60, 61, 15, 62].

Limits of the present data. All doses so far were made in air, on one auger geometry, and the optimization covers one powder at one target mass. Barium chloride caked after humidity cycling, and the correlation of dose error with moisture content shows that ambient humidity matters for hygroscopic powders. The tap stage sets the cost of accuracy. All 16 validation doses landed inside the 3 mg band, but taps took most of each dose's time, and the mass a tap frees falls as the lip of the tube empties (on salt, from about 2.8 mg for the first tap after the auger moved to below 0.5 mg after 40 taps). Tighter tolerances or faster doses therefore need a smaller and more consistent final increment, for example a smaller outlet or a refill of the lip between taps, more than a different control law [64]. The feed factor also drifted between sessions, by about 10% for salt between two days and up to 1.8-fold across sessions with different fill levels, so a fingerprint needs a standardized powder state (fill level, priming and conditioning) and a same-day reference powder before small differences between batches can be read from it. Finally, every powder was supplied commercially or made in our laboratory without a recorded lot analysis, which is why measured properties of each lot are part of the work that remains.

Implications for additive manufacturing. In an alloy-discovery workflow the doser is calibrated on every powder anyway, so the records described here come at little extra cost in powder or time. Stored with the dose records, the fingerprint, the validated settings and the inferred contact parameters for each lot form a feedstock record that can flag a batch that flows unlike its predecessors, as the clogged Al 4047 batch would have been flagged before it reached a printer. [Whether that record predicts spreading and packing in the powder bed will be tested on blade-spread layers (Section 2.7).]

5. Conclusions

We treated the calibration of an open-source auger doser as multi-objective Bayesian optimization and asked what the calibration data say about the powder. On salt, an unattended campaign of 42 doses found settings that dosed 0.5 g in a median 78 s with a median error of 1.8 mg over eight repeats,

three times faster than hand-tuned settings, and those settings dosed 8 g of AlSi10Mg to within 0.1% without change in a single transfer check. At fixed settings, 13 powders spanned a 24-fold range in delivery per revolution, three did not flow at all, and the delivery ranked powders like their angle of repose. A discrete element model of the auger, with no parameters fitted to dosing data, reproduced the measured delivery of salt at one setting and its trend with tilt. [Multi-task calibration across N powders reduced the doses needed for a new powder by X%, and effective cohesion and friction inferred from dosing data tracked shear-cell and Hall-flow measurements.] Because calibration is a step every dosing workflow already performs, its records can double as a characterization of each feedstock at little extra cost.

CRediT authorship contribution statement

[To be completed once the author list is agreed.]

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data, code, CAD files and simulation inputs are available at `https://github.com/vertical-cloud-lab/powder-doser` [and will be archived with a DOI on acceptance].

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude Code (Anthropic) in order to draft and revise the text and to write the figure scripts, and Edison Scientific in order to search the literature and to obtain mock reviews of the draft. After using these tools/services, the authors

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References

[1] M. H. Mosallanejad, B. Niroumand, A. Aversa, A. Saboori, In-situ alloying in laser-based additive manufacturing processes: A critical review, Journal of Alloys and Compounds 872 (2021) 159567. doi:10.1016/j.jallcom.2021.159567.
URL http://dx.doi.org/10.1016/j.jallcom.2021.159567

[2] M. Moorehead, K. Bertsch, M. Niezgoda, C. Parkin, M. Elbakhshwan, K. Sridharan, C. Zhang, D. Thoma, A. Couet, High-throughput synthesis of mo-nb-ta-w high-entropy alloys via additive manufacturing, Materials & Design 187 (2020) 108358. doi:10.1016/j.matdes.2019.108358.
URL http://dx.doi.org/10.1016/j.matdes.2019.108358

[3] Z. Li, A. Ludwig, A. Savan, H. Springer, D. Raabe, Combinatorial metallurgical synthesis and processing of high-entropy alloys, Journal of Materials Research 33 (2018) 3156–3169. doi:10.1557/jmr.2018.214.
URL https://doi.org/10.1557/jmr.2018.214

[4] K. S. Vecchio, O. F. Dippo, K. R. Kaufmann, X. Liu, High-throughput rapid experimental alloy development (ht-read), Acta Materialia 221 (2021) 117352. doi:10.1016/j.actamat.2021.117352.
URL https://doi.org/10.1016/j.actamat.2021.117352

[5] S. Mooraj, G. Kim, X. Fan, S. Samuha, Y. Xie, T. Li, J. S. Tiley, Y. Chen, D. Yu, K. An, P. Hosemann, P. K. Liaw, W. Chen, W. Chen, Additive manufacturing of defect-free tizrnbta refractory high-entropy alloy with enhanced elastic isotropy via in-situ alloying of elemental powders, Communications Materials 5 (2024) 14. doi:10.1038/s43246-024-00452-0.
URL https://doi.org/10.1038/s43246-024-00452-0

[6] P. Nelaturu, J. R. Hattrick-Simpers, M. Moorehead, V. Jambur, I. Szlufarska, A. Couet, D. J. Thoma, Multi-principal element alloy discovery using directed energy deposition and machine learning, Materials Science and Engineering: A 891 (2024) 145945. doi:10.1016/

j.msea.2023.145945.
URL https://doi.org/10.1016/j.msea.2023.145945

[7] M. Abolhasani, E. Kumacheva, The rise of self-driving labs in chemical and materials sciences, Nature Synthesis 2 (6) (2023) 483–492. doi:10.1038/s44160-022-00231-0.
URL http://dx.doi.org/10.1038/s44160-022-00231-0

[8] G. Tom, S. P. Schmid, S. G. Baird, Y. Cao, K. Darvish, H. Hao, S. Lo, S. Pablo-García, E. M. Rajaonson, M. Skreta, N. Yoshikawa, S. Corapi, G. D. Akkoc, F. Strieth-Kalthoff, M. Seifrid, A. Aspuru-Guzik, Self-driving laboratories for chemistry and materials science, Chemical Reviews 124 (16) (2024) 9633–9732. doi:10.1021/acs.chemrev.4c00055.
URL http://dx.doi.org/10.1021/acs.chemrev.4c00055

[9] M. Seifrid, R. Pollice, A. Aguilar-Granda, Z. M. Chan, K. Hotta, C. T. Ser, J. Vestfrid, T. C. Wu, A. Aspuru-Guzik, Autonomous chemical experiments: Challenges and perspectives on establishing a self-driving lab, Accounts of Chemical Research 55 (2022) 2454–2466. doi:10.1021/acs.accounts.2c00220.
URL https://doi.org/10.1021/acs.accounts.2c00220

[10] N. J. Szymanski, B. Rendy, Y. Fei, R. E. Kumar, T. He, D. Milsted, M. J. McDermott, M. Gallant, E. D. Cubuk, A. Merchant, H. Kim, A. Jain, C. J. Bartel, K. Persson, Y. Zeng, G. Ceder, An autonomous laboratory for the accelerated synthesis of inorganic materials, Nature 624 (7990) (2023) 86–91. doi:10.1038/s41586-023-06734-w.
URL http://dx.doi.org/10.1038/s41586-023-06734-w

[11] A. M. Lunt, H. Fakhruldeen, G. Pizzuto, L. Longley, A. White, N. Rankin, R. Clowes, B. Alston, L. Gigli, G. M. Day, A. I. Cooper, S. Y. Chong, Modular, multi-robot integration of laboratories: an autonomous workflow for solid-state chemistry, Chemical Science 15 (7) (2024) 2456–2463. doi:10.1039/d3sc06206f.
URL http://dx.doi.org/10.1039/D3SC06206F

[12] Y. Jiang, H. Fakhruldeen, G. Pizzuto, L. Longley, A. He, T. Dai, R. Clowes, N. Rankin, A. I. Cooper, Autonomous biomimetic solid dispensing using a dual-arm robotic manipulator, Digital Discovery 2 (6)

(2023) 1733–1744. doi:10.1039/d3dd00075c.
URL http://dx.doi.org/10.1039/D3DD00075C

[13] P. M. Maffettone, P. Friederich, S. G. Baird, B. Blaiszik, K. A. Brown, S. I. Campbell, O. A. Cohen, R. L. Davis, I. T. Foster, N. Haghmoradi, M. Hereld, H. Joress, N. Jung, H.-K. Kwon, G. Pizzuto, J. Rintamaki, C. Steinmann, L. Torresi, S. Sun, What is missing in autonomous discovery: open challenges for the community, Digital Discovery 2 (2023) 1644–1659, nEEDS MANUAL VERIFICATION: DOI returned non-404 but Crossref metadata unavailable. doi:10.48550/arxiv.2304.11120.
URL https://doi.org/10.48550/arxiv.2304.11120

[14] S. Vock, B. Klöden, A. Kirchner, T. Weißgärber, B. Kieback, Powders for powder bed fusion: a review, Progress in Additive Manufacturing 4 (4) (2019) 383–397. doi:10.1007/s40964-019-00078-6.
URL http://dx.doi.org/10.1007/s40964-019-00078-6

[15] C. Meier, R. Weissbach, J. Weinberg, W. A. Wall, A. J. Hart, Critical influences of particle size and adhesion on the powder layer uniformity in metal additive manufacturing, Journal of Materials Processing Technology 266 (2019) 484–501. doi:10.1016/j.jmatprotec.2018.10.037.
URL http://dx.doi.org/10.1016/j.jmatprotec.2018.10.037

[16] S. E. Brika, M. Letenneur, C. A. Dion, V. Brailovski, Influence of particle morphology and size distribution on the powder flowability and laser powder bed fusion manufacturability of ti-6al-4v alloy, Additive Manufacturing 31 (2020) 100929. doi:10.1016/j.addma.2019.100929.
URL http://dx.doi.org/10.1016/j.addma.2019.100929

[17] L. Haferkamp, L. Haudenschild, A. Spierings, K. Wegener, K. Riener, S. Ziegelmeier, G. J. Leichtfried, The influence of particle shape, powder flowability, and powder layer density on part density in laser powder bed fusion, Metals 11 (3) (2021) 418. doi:10.3390/met11030418.
URL http://dx.doi.org/10.3390/met11030418

[18] A. B. Spierings, M. Voegtlin, T. Bauer, K. Wegener, Powder flowability characterisation methodology for powder-bed-based metal additive manufacturing, Progress in Additive Manufacturing 1 (1-2) (2015) 9–20.

doi:10.1007/s40964-015-0001-4.
URL http://dx.doi.org/10.1007/s40964-015-0001-4

[19] M. Krantz, H. Zhang, J. Zhu, Characterization of powder flow: Static and dynamic testing, Powder Technology 194 (3) (2009) 239–245. doi:10.1016/j.powtec.2009.05.001.
URL http://dx.doi.org/10.1016/j.powtec.2009.05.001

[20] D. Schulze, Powders and Bulk Solids: Behavior, Characterization, Storage and Flow, Springer International Publishing, 2021. doi:10.1007/978-3-030-76720-4.
URL http://dx.doi.org/10.1007/978-3-030-76720-4

[21] R. Freeman, Measuring the flow properties of consolidated, conditioned and aerated powders — a comparative study using a powder rheometer and a rotational shear cell, Powder Technology 174 (1-2) (2007) 25–33. doi:10.1016/j.powtec.2006.10.016.
URL http://dx.doi.org/10.1016/j.powtec.2006.10.016

[22] G. Lumay, F. Boschini, K. Traina, S. Bontempi, J.-C. Remy, R. Cloots, N. Vandewalle, Measuring the flowing properties of powders and grains, Powder Technology 224 (2012) 19–27. doi:10.1016/j.powtec.2012.02.015.
URL http://dx.doi.org/10.1016/j.powtec.2012.02.015

[23] G. Yablokova, M. Speirs, J. Van Humbeeck, J.-P. Kruth, J. Schrooten, R. Cloots, F. Boschini, G. Lumay, J. Luyten, Rheological behavior of β-ti and niti powders produced by atomization for slm production of open porous orthopedic implants, Powder Technology 283 (2015) 199–209. doi:10.1016/j.powtec.2015.05.015.
URL http://dx.doi.org/10.1016/j.powtec.2015.05.015

[24] Z. Snow, R. Martukanitz, S. Joshi, On the development of powder spreadability metrics and feedstock requirements for powder bed fusion additive manufacturing, Additive Manufacturing 28 (2019) 78–86. doi:10.1016/j.addma.2019.04.017.
URL http://dx.doi.org/10.1016/j.addma.2019.04.017

[25] L. Cordova, T. Bor, M. de Smit, M. Campos, T. Tinga, Measuring the spreadability of pre-treated and moisturized powders for laser powder

bed fusion, Additive Manufacturing 32 (2020) 101082. doi:10.1016/j.addma.2020.101082.
URL http://dx.doi.org/10.1016/j.addma.2020.101082

[26] Y. Wang, T. Li, F. J. Muzzio, B. J. Glasser, Predicting feeder performance based on material flow properties, Powder Technology 308 (2017) 135–148. doi:10.1016/j.powtec.2016.12.010.
URL http://dx.doi.org/10.1016/j.powtec.2016.12.010

[27] F. Tahir, J. Palmer, J. Khoo, J. Holman, I. K. Yadav, G. Reynolds, E. Meehan, A. Mitchell, G. Bajwa, Development of feed factor prediction models for loss-in-weight powder feeders, Powder Technology 364 (2020) 1025–1038. doi:10.1016/j.powtec.2019.09.071.
URL http://dx.doi.org/10.1016/j.powtec.2019.09.071

[28] B. Bekaert, L. Penne, W. Grymonpré, B. Van Snick, J. Dhondt, J. Boeckx, J. Vogeleer, T. De Beer, C. Vervaet, V. Vanhoorne, Determination of a quantitative relationship between material properties, process settings and screw feeding behavior via multivariate data-analysis, International Journal of Pharmaceutics 602 (2021) 120603. doi:10.1016/j.ijpharm.2021.120603.
URL http://dx.doi.org/10.1016/j.ijpharm.2021.120603

[29] W. E. Engisch, F. J. Muzzio, Method for characterization of loss-in-weight feeder equipment, Powder Technology 228 (2012) 395–403. doi:10.1016/j.powtec.2012.05.058.
URL http://dx.doi.org/10.1016/j.powtec.2012.05.058

[30] W. E. Engisch, F. J. Muzzio, Feedrate deviations caused by hopper refill of loss-in-weight feeders, Powder Technology 283 (2015) 389–400. doi:10.1016/j.powtec.2015.06.001.
URL https://doi.org/10.1016/j.powtec.2015.06.001

[31] J. Hanson, Control of a system of loss-in-weight feeders for drug product continuous manufacturing, Powder Technology 331 (2018) 236–243. doi:10.1016/j.powtec.2018.03.027.
URL https://doi.org/10.1016/j.powtec.2018.03.027

[32] M. Besenhard, E. Faulhammer, S. Fathollahi, G. Reif, V. Calzolari, S. Biserni, A. Ferrari, S. Lawrence, M. Llusa, J. Khinast, Accuracy

of micro powder dosing via a vibratory sieve–chute system, European Journal of Pharmaceutics and Biopharmaceutics 94 (2015) 264–272. doi:10.1016/j.ejpb.2015.04.037.
URL https://doi.org/10.1016/j.ejpb.2015.04.037

[33] M. Besenhard, S. Fathollahi, E. Siegmann, E. Slama, E. Faulhammer, J. Khinast, Micro-feeding and dosing of powders via a small-scale powder pump, International Journal of Pharmaceutics 519 (1-2) (2017) 314–322. doi:10.1016/j.ijpharm.2016.12.029.
URL http://dx.doi.org/10.1016/j.ijpharm.2016.12.029

[34] E. Faulhammer, M. Fink, M. Llusa, S. M. Lawrence, S. Biserni, V. Calzolari, J. G. Khinast, Low-dose capsule filling of inhalation products: Critical material attributes and process parameters, International Journal of Pharmaceutics 473 (2014) 617–626. doi:10.1016/j.ijpharm.2014.07.050.
URL https://doi.org/10.1016/j.ijpharm.2014.07.050

[35] S. Yang, J. Evans, Metering and dispensing of powder; the quest for new solid freeforming techniques, Powder Technology 178 (2007) 56–72. doi:10.1016/j.powtec.2007.04.004.
URL https://doi.org/10.1016/j.powtec.2007.04.004

[36] J. Alsenz, Powderpicking: An inexpensive, manual, medium-throughput method for powder dispensing, Powder Technology 209 (1-3) (2011) 152–157. doi:10.1016/j.powtec.2011.02.014.
URL http://dx.doi.org/10.1016/j.powtec.2011.02.014

[37] B. Shahriari, K. Swersky, Z. Wang, R. P. Adams, N. de Freitas, Taking the human out of the loop: A review of bayesian optimization, Proceedings of the IEEE 104 (1) (2016) 148–175. doi:10.1109/jproc.2015.2494218.
URL http://dx.doi.org/10.1109/JPROC.2015.2494218

[38] Q. Liang, A. E. Gongora, Z. Ren, A. Tiihonen, Z. Liu, S. Sun, J. R. Deneault, D. Bash, F. Mekki-Berrada, S. A. Khan, K. Hippalgaonkar, B. Maruyama, K. A. Brown, J. Fisher III, T. Buonassisi, Benchmarking the performance of bayesian optimization across multiple experimental materials science domains, npj Computational Materials 7 (1) (2021). doi:10.1038/s41524-021-00656-9.
URL http://dx.doi.org/10.1038/s41524-021-00656-9

[39] S. G. Baird, A. R. Falkowski, T. D. Sparks, Honegumi: an interface for accelerating the adoption of Bayesian optimization in the experimental sciences, npj Computational Materials 12 (2026) 296. doi:10.1038/s41524-026-02156-0.

[40] J. R. Deneault, J. Chang, J. Myung, D. Hooper, A. Armstrong, M. Pitt, B. Maruyama, Toward autonomous additive manufacturing: Bayesian optimization on a 3d printer, MRS Bulletin 46 (7) (2021) 566–575. doi:10.1557/s43577-021-00051-1.
URL http://dx.doi.org/10.1557/s43577-021-00051-1

[41] S. Daulton, M. Balandat, E. Bakshy, Parallel Bayesian optimization of multiple noisy objectives with expected hypervolume improvement, in: Advances in Neural Information Processing Systems, Vol. 34, 2021, pp. 2187–2200. arXiv:2105.08195.

[42] D. Eriksson, M. Jankowiak, High-dimensional Bayesian optimization with sparse axis-aligned subspaces, in: Proceedings of the Thirty-Seventh Conference on Uncertainty in Artificial Intelligence, Vol. 161 of Proceedings of Machine Learning Research, 2021, pp. 493–503. arXiv:2103.00349.

[43] E. V. Bonilla, K. M. A. Chai, C. K. I. Williams, Multi-task Gaussian process prediction, in: Advances in Neural Information Processing Systems, Vol. 20, 2008, pp. 153–160.
URL https://papers.nips.cc/paper/3189-multi-task-gaussian-process-prediction

[44] K. Swersky, J. Snoek, R. P. Adams, Multi-task Bayesian optimization, in: Advances in Neural Information Processing Systems, Vol. 26, 2013, pp. 2004–2012.
URL https://papers.nips.cc/paper/5086-multi-task-bayesian-optimization

[45] C. J. Taylor, K. C. Felton, D. Wigh, M. I. Jeraal, R. Grainger, G. Chessari, C. N. Johnson, A. A. Lapkin, Accelerated chemical reaction optimization using multi-task learning, ACS Central Science 9 (5) (2023) 957–968. doi:10.1021/acscentsci.3c00050.
URL http://dx.doi.org/10.1021/acscentsci.3c00050

[46] P. A. Cundall, O. D. L. Strack, A discrete numerical model for granular assemblies, Géotechnique 29 (1) (1979) 47–65. doi:10.1680/geot.1979.29.1.47.
URL http://dx.doi.org/10.1680/geot.1979.29.1.47

[47] H. Zhu, Z. Zhou, R. Yang, A. Yu, Discrete particle simulation of particulate systems: A review of major applications and findings, Chemical Engineering Science 63 (23) (2008) 5728–5770. doi:10.1016/j.ces.2008.08.006.
URL http://dx.doi.org/10.1016/j.ces.2008.08.006

[48] P. Owen, P. Cleary, Prediction of screw conveyor performance using the discrete element method (dem), Powder Technology 193 (3) (2009) 274–288. doi:10.1016/j.powtec.2009.03.012.
URL http://dx.doi.org/10.1016/j.powtec.2009.03.012

[49] K. L. Johnson, K. Kendall, A. D. Roberts, Surface energy and the contact of elastic solids, Proceedings of the Royal Society of London. A. Mathematical and Physical Sciences 324 (1558) (1971) 301–313. doi:10.1098/rspa.1971.0141.
URL http://dx.doi.org/10.1098/rspa.1971.0141

[50] S. C. Thakur, J. P. Morrissey, J. Sun, J. F. Chen, J. Y. Ooi, Micromechanical analysis of cohesive granular materials using the discrete element method with an adhesive elasto-plastic contact model, Granular Matter 16 (3) (2014) 383–400. doi:10.1007/s10035-014-0506-4.
URL http://dx.doi.org/10.1007/s10035-014-0506-4

[51] J. Ai, J.-F. Chen, J. M. Rotter, J. Y. Ooi, Assessment of rolling resistance models in discrete element simulations, Powder Technology 206 (3) (2011) 269–282. doi:10.1016/j.powtec.2010.09.030.
URL http://dx.doi.org/10.1016/j.powtec.2010.09.030

[52] C. Coetzee, Review: Calibration of the discrete element method, Powder Technology 310 (2017) 104–142. doi:10.1016/j.powtec.2017.01.015.
URL http://dx.doi.org/10.1016/j.powtec.2017.01.015

[53] C. Richter, T. Rößler, G. Kunze, A. Katterfeld, F. Will, Development of a standard calibration procedure for the dem parameters of cohesionless bulk materials – part ii: Efficient optimization-based calibration, Powder

Technology 360 (2020) 967–976. doi:10.1016/j.powtec.2019.10.052.
URL http://dx.doi.org/10.1016/j.powtec.2019.10.052

[54] L. Benvenuti, C. Kloss, S. Pirker, Identification of dem simulation parameters by artificial neural networks and bulk experiments, Powder Technology 291 (2016) 456–465. doi:10.1016/j.powtec.2016.01.003.
URL http://dx.doi.org/10.1016/j.powtec.2016.01.003

[55] H. Cheng, T. Shuku, K. Thoeni, P. Tempone, S. Luding, V. Magnanimo, An iterative bayesian filtering framework for fast and automated calibration of dem models, Computer Methods in Applied Mechanics and Engineering 350 (2019) 268–294. doi:10.1016/j.cma.2019.01.027.
URL http://dx.doi.org/10.1016/j.cma.2019.01.027

[56] M. C. Kennedy, A. O'Hagan, Bayesian calibration of computer models, Journal of the Royal Statistical Society Series B: Statistical Methodology 63 (3) (2001) 425–464. doi:10.1111/1467-9868.00294.
URL http://dx.doi.org/10.1111/1467-9868.00294

[57] D. Higdon, J. Gattiker, B. Williams, M. Rightley, Computer model calibration using high-dimensional output, Journal of the American Statistical Association 103 (482) (2008) 570–583. doi:10.1198/016214507000000888.
URL http://dx.doi.org/10.1198/016214507000000888

[58] B. E. Kassem, N. Salloum, T. Brinz, Y. Heider, B. Markert, A semi-automated dem parameter calibration technique of powders based on different bulk responses extracted from auger dosing experiments, KONA Powder and Particle Journal 38 (0) (2021) 235–250. doi:10.14356/kona.2021010.
URL http://dx.doi.org/10.14356/kona.2021010

[59] B. El-Kassem, N. Salloum, T. Brinz, Y. Heider, B. Markert, A multivariate regression parametric study on dem input parameters of free-flowing and cohesive powders with experimental data-based validation, Computational Particle Mechanics 8 (1) (2021) 87–111. doi:10.1007/s40571-020-00315-8.
URL http://dx.doi.org/10.1007/s40571-020-00315-8

[60] S. Haeri, Optimisation of blade type spreaders for powder bed preparation in additive manufacturing using dem simulations, Powder Technology 321 (2017) 94–104. doi:10.1016/j.powtec.2017.08.011.
URL http://dx.doi.org/10.1016/j.powtec.2017.08.011

[61] W. Nan, M. Pasha, T. Bonakdar, A. Lopez, U. Zafar, S. Nadimi, M. Ghadiri, Jamming during particle spreading in additive manufacturing, Powder Technology 338 (2018) 253–262. doi:10.1016/j.powtec.2018.07.030.
URL http://dx.doi.org/10.1016/j.powtec.2018.07.030

[62] M. Y. Shaheen, A. R. Thornton, S. Luding, T. Weinhart, The influence of material and process parameters on powder spreading in additive manufacturing, Powder Technology 383 (2021) 564–583. doi:10.1016/j.powtec.2021.01.058.
URL http://dx.doi.org/10.1016/j.powtec.2021.01.058

[63] S. Charles, W. Mulberry, L. Winters, C. Robison, G. Erickson, S. G. Baird, An open-source, 3D-printed auger powder doser designed with generative artificial intelligence, manuscript in preparation (2026).

[64] Vertical Cloud Lab, powder-doser: an open-source auger powder doser for self-driving laboratories, https://github.com/vertical-cloud-lab/powder-doser (2026).

[65] T. R. Brocksieper, T. Grimm, J. T. Sehrt, Investigation of flowability and spreadability of metal powders for powder bed fusion additive manufacturing, Progress in Additive Manufacturing 11 (7) (2026) 6661–6673. doi:10.1007/s40964-026-01690-z.
URL http://dx.doi.org/10.1007/s40964-026-01690-z

[66] Meta Platforms, Inc., Ax: Adaptive experimentation platform, https://ax.dev, version 0.4.3 (2024).

[67] M. Balandat, B. Karrer, D. R. Jiang, S. Daulton, B. Letham, A. G. Wilson, E. Bakshy, BoTorch: A framework for efficient Monte-Carlo Bayesian optimization, in: Advances in Neural Information Processing Systems, Vol. 33, 2020, pp. 21524–21538. arXiv:1910.06403.

[68] S. Ament, S. Daulton, D. Eriksson, M. Balandat, E. Bakshy, Unexpected improvements to expected improvement for bayesian optimization,

in: Advances in Neural Information Processing Systems 36, Neural Information Processing Systems Foundation, Inc. (NeurIPS), 2023, pp. 20577–20612. doi:10.52202/075280-0904.
URL http://dx.doi.org/10.52202/075280-0904

[69] C. Kloss, C. Goniva, A. Hager, S. Amberger, S. Pirker, Models, algorithms and validation for opensource dem and cfd-dem, Progress in Computational Fluid Dynamics, An International Journal 12 (2/3) (2012) 140. doi:10.1504/pcfd.2012.047457.
URL http://dx.doi.org/10.1504/PCFD.2012.047457

[70] S. Girnth, T. Heitkamp, C. Wacker, N. Waldt, G. Klawitter, K. Dröder, Dimensionless quantities in discrete element method: powder model parameterization for additive manufacturing, Progress in Additive Manufacturing 9 (6) (2024) 1967–1983. doi:10.1007/s40964-023-00543-3.
URL http://dx.doi.org/10.1007/s40964-023-00543-3

[71] ASTM International, ASTM F3049-14: Standard guide for characterizing properties of metal powders used for additive manufacturing processes, ASTM International, West Conshohocken, PA (2014).

[72] ASTM International, ASTM B822-20: Standard test method for particle size distribution of metal powders and related compounds by light scattering, ASTM International, West Conshohocken, PA (2020).

[73] ASTM International, ASTM B527-22: Standard test method for tap density of metal powders and compounds, ASTM International, West Conshohocken, PA (2022).

[74] ASTM International, ASTM B213-20: Standard test methods for flow rate of metal powders using the Hall flowmeter funnel, ASTM International, West Conshohocken, PA (2020).

[75] ASTM International, ASTM B964-16: Standard test methods for flow rate of metal powders using the Carney funnel, ASTM International, West Conshohocken, PA (2016).

[76] ASTM International, ASTM D7891-15: Standard test method for shear testing of powders using the Freeman technology FT4 powder rheometer shear cell, ASTM International, West Conshohocken, PA (2015).

[77] L. Marchetti, C. Hulme-Smith, Flowability of steel and tool steel powders: A comparison between testing methods, Powder Technology 384 (2021) 402–413. doi:10.1016/j.powtec.2021.01.074.
URL http://dx.doi.org/10.1016/j.powtec.2021.01.074

[78] N. Bostijn, J. Dhondt, A. Ryckaert, E. Szabó, W. Dhondt, B. Van Snick, V. Vanhoorne, C. Vervaet, T. De Beer, A multivariate approach to predict the volumetric and gravimetric feeding behavior of a low feed rate feeder based on raw material properties, International Journal of Pharmaceutics 557 (2019) 342–353. doi:10.1016/j.ijpharm.2018.12.066.
URL http://dx.doi.org/10.1016/j.ijpharm.2018.12.066

[79] E. Galindo, E. R. L. Espiritu, C. Gutierrez, A. N. Alagha, P. Hudon, M. Brochu, A method to assess the quality of additive manufacturing metal powders using the triboelectric charging concept, Scientific Reports 14 (1) (2024). doi:10.1038/s41598-024-67295-0.
URL http://dx.doi.org/10.1038/s41598-024-67295-0

Table 2: Descriptors of the dosing fingerprint. "Measured" means available from existing records for the number of powders given; "exploratory" means computed once from existing records with a protocol that has not been repeated; "planned" means not yet measured.

Descriptor	Unit	Source	Replicates	Status
Delivery per revolution at 0, 22.5, 45°	mg/rev	protocol C, 30 rpm	6 rev per tilt	measured, 13
Revolution-to-revolution scatter (CoV)	%	protocol C	6 rev per tilt	measured, 10
Speed exponent α	–	protocol D, 15/45/90 rpm	3 rev per speed	measured, 8
Mass freed by one tap	mg/tap	protocol E	8 taps	measured where resolved, 7
Afterflow time constant τ_{af}	s	stops in closed-loop doses	all stops of a run	measured, 6
Interruption fraction	–	15 rpm balance increments	1 run	exploratory, 10
Volumetric flow index	–	45° delivery / bulk density	20 rev	planned

Table 3: Search space of the salt campaign (target 0.5 g). The two tap settings are categorical and the rest continuous. The hand-tuned settings are those found by hand for this controller on salt before the campaign; the recommended settings are dose `bo-005`.

Setting	Range	Hand-tuned	Recommended
Taps during bulk stage	off / 2 Hz	off	2 Hz
Taps during trickle stage	off / 2 Hz	off	off
Bulk tilt (°)	15–40	30	40
Bulk speed (rpm)	20–100	55	100
Switch to trickle, mass to go (g)	0.05–0.30	0.25	0.30
Trickle tilt (°)	10–30	15	10
Tap tilt (°)	0–15	10	15
Stopping band $\pm\delta$ (mg)	3–15	5	3

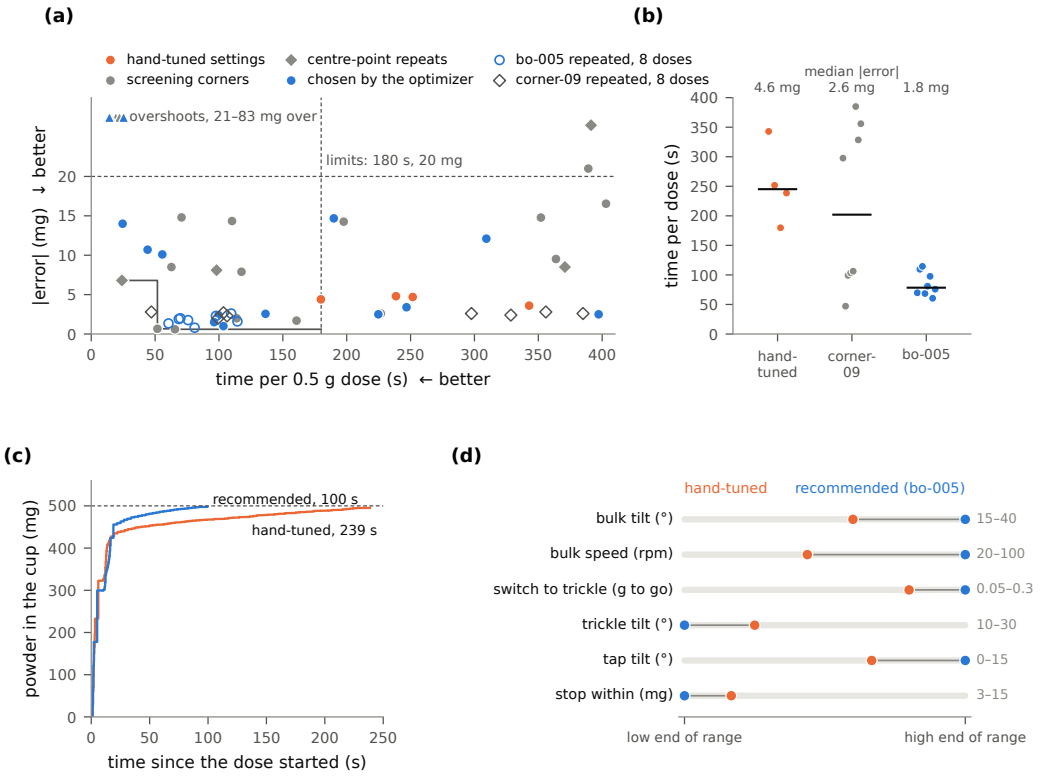


Figure 3: Multi-objective Bayesian calibration of the doser on salt (target 0.5 g). (a) Time and absolute error of every dose; triangles at the top mark overshoots beyond the 20 mg limit, the dashed lines are the limits that define the hypervolume, and the step line is the observed Pareto front. Open markers are the two validation blocks of eight doses. (b) Time per dose for the hand-tuned settings (4 doses) and the two validated settings (8 doses each); bars are medians, and the numbers above give the median absolute error. (c) Powder in the cup during a hand-tuned dose and a dose at the recommended settings. (d) The six numeric settings, each drawn on its own search range, for the hand-tuned and recommended settings.

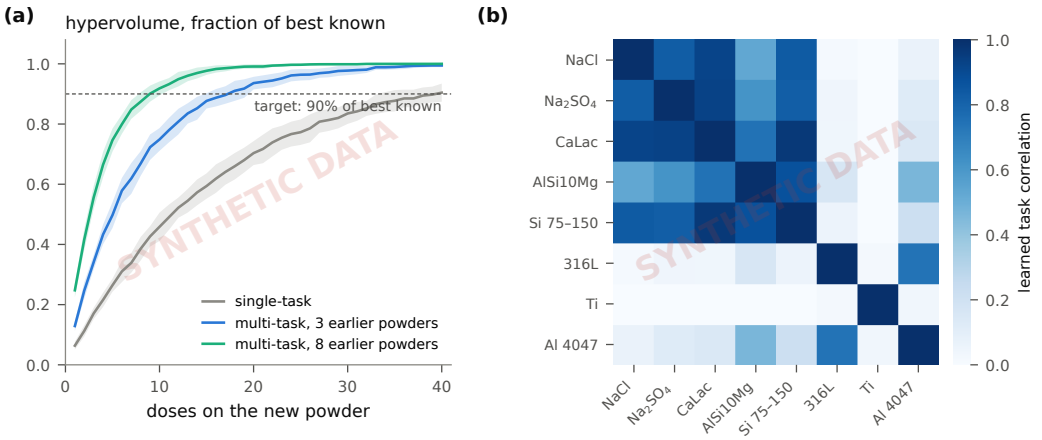


Figure 4: [Placeholder: synthetic data.] Multi-task calibration across powders. (a) Hypervolume reached on a held-out powder against the number of doses on that powder, for single-task optimization and for multi-task optimization informed by three or eight earlier powders (mean ± one standard deviation over ten replays). (b) Correlations between powders learned by the multi-task model. All numbers in this figure are invented to show the form of the result.

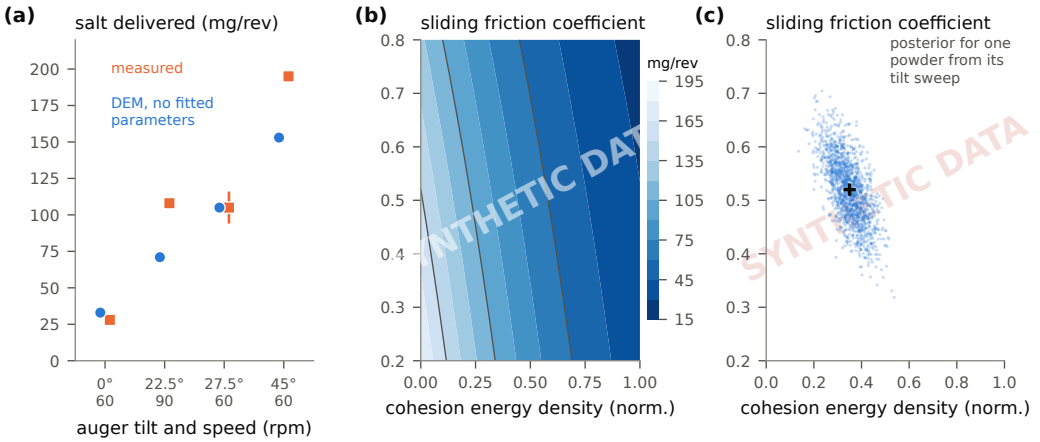


Figure 5: Discrete element model of the auger. (a) Salt delivered per revolution, simulated with no parameters fitted to dosing data and measured on the rig, at four combinations of tilt and speed. At 27.5° and 60 rpm the measurement is the campaign's centre point (eight doses, bar: ± one standard deviation); at 22.5° both simulation and measurement are at 90 rpm (bar: range of runs); the 0° and 45° measurements were made at 30 rpm and scaled to the simulated 60 rpm with the measured speed exponent. (b) [Placeholder: synthetic data.] Delivery per revolution against effective cohesion and friction from the emulator. (c) [Placeholder: synthetic data.] Posterior distribution of the two parameters for one powder given its tilt sweep. Panels (b) and (c) show invented numbers to illustrate the form of the result.

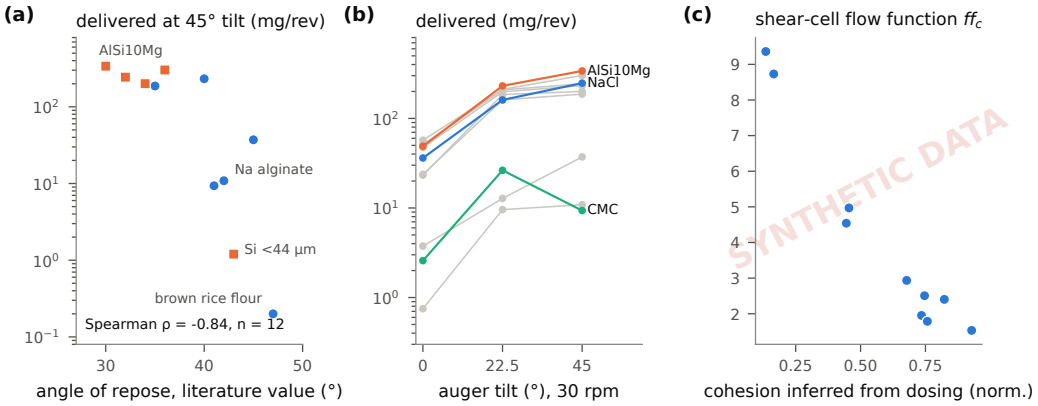


Figure 6: Dosing response against powder properties. (a) Delivery per revolution at 45° tilt and 30 rpm against the literature angle of repose for the material class (seven of twelve values are engineering estimates); the finer silicon grade (one 7 mg pulse in six revolutions, 1.2 mg per revolution) and brown rice flour (0.2 mg per revolution) are plotted at their means, which are not resolved from zero. Blue circles: food-safe stand-ins; orange squares: alloy-relevant powders. (b) Delivery against tilt for the ten conveying powders; three are highlighted. (c) [Placeholder: synthetic data.] Effective cohesion inferred from dosing against the shear-cell flow function coefficient measured on the same lots.